

School of Engineering and Technology, MGM University Chh. Sambhajinagar

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Mini Project report
On

"STUDENT MANAGEMENT SYSTEM"

By

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CERTIFICATE

This to certify that **Student ADITYA SANAP , ABHIJEET PATIL** with PRN No.**, 202456108013,2024561012** has successfully completed their Mini-Project entitled "**STUDENT MANAGEMENT SYSTEM** "the Course of **"Java Programming"** in the fifth Semester during tenure of completing the B. Tech program in **Computer Science and Engineering (IoT,Cyber Security Including Blockchain Technolpgy** from **School of Engineering and Technology, MGM University** institute.

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1. Abstract

The Student Management System is a software application designed to organize and manage student information in an educational institution. Maintaining student records manually can take time, create duplicate entries, and make it difficult to update or retrieve information. This project provides a structured way to manage student details such as student ID, name, department, semester, contact information, and academic records.

The system is intended to support common operations such as adding new student records, viewing the list of students, searching for a student, updating details, and deleting records when required. The application demonstrates how software can improve the accuracy, accessibility, and organization of student information. It also provides practical experience in requirement analysis, user interface design, data handling, implementation, and software testing.

The exact technology stack can be adapted to the implementation used by the project team. The report describes a typical student record management application and should be updated where necessary to match the final working software.

2. Introduction

Educational institutions maintain information about students for academic and administrative purposes. These records may include personal details, roll numbers, departments, semesters,

contact details, and marks. When this information is maintained only in paper registers or separate files, retrieving and updating records can be inconvenient.
The Student Management System aims to provide a computerized solution for maintaining student records in an organized manner. Authorized users can enter student details, view stored records, search for a particular student, and update or remove information when necessary. A structured system can reduce repetitive manual work and make record management more consistent.

Objectives

To maintain student information in an organized format.
To provide options to add, view, search, update, and delete student records.
To reduce manual paperwork and duplicate record entry.
To make student information easier to retrieve and maintain.
To apply software engineering practices during development and testing.

3. Course Outcomes Integrated

Sr. No.	Course Outcome	Application in the Project
1	Requirement Analysis	Identify the information and operations required by student record users.
2	System Design	Plan the application modules, data fields, and user interface.
3	Software Development	Implement the functions needed to manage student records.
4	Data Management	Store, retrieve, and update student details using the chosen storage method.
5	Software Testing	Test adding, viewing, searching, editing, and deleting records.
6	Maintenance	Identify improvements and correct defects found during use.

4. Action Plan

S. No.	Activity Details	Planned Start	Planned Finish	Responsible Members
1	Select the project topic and define objectives	01/10/2026	02/10/2026	Project Team
2	Gather and analyse requirements	03/10/2026	04/10/2026	Project Team
3	Design the system and student data structure	05/10/2026	07/10/2026	Project Team
4	Develop the interface and core functions	08/10/2026	12/10/2026	Project Team
5	Integrate data storage and test functions	13/10/2026	15/10/2026	Project Team
6	Correct errors and prepare the	16/10/2026	18/10/2026	Project Team

	report			

Note: Update the dates and team-member names to match the actual project work.

5. Literature Review

Traditional Record Keeping

In traditional methods, student information is recorded in registers, paper forms, or separate spreadsheets. These methods are easy to begin with but may require extra effort to search, update, and maintain records consistently.

Computerized Student Records

Computerized record systems allow student details to be entered and retrieved electronically. Search and update operations can be performed more quickly than searching through paper records, provided the data is entered correctly and access is managed appropriately.

Structured Data Storage

A structured data store helps keep student information in a consistent format. Fields such as student ID, name, department, semester, email, and phone number can be maintained as separate attributes. Validation can reduce incomplete or incorrectly formatted entries.

Proposed System

The proposed Student Management System brings common student record operations into one application. It is intended to simplify data entry and retrieval, reduce manual effort, and provide a foundation for future features such as attendance, marks management, authentication, and report generation.

6. Data Flow Diagram / Use Case Diagram

A. Data Flow Diagram (DFD) – Level 0

The DFD describes how information moves between the user, the Student Management System, and the student records store.

Entity / Process	Description
Student / Staff / Administrator	Provides student information or requests a record.
Student Management System	Validates input and processes add, view, search, update, and delete requests.
Student Records Store	Stores student records and returns matching information.
Output	Displays saved student details, search results, or operation messages.

Flow: User submits a request → System validates and processes the request → Student record store is read or updated → System displays the result.

B. Use Case Diagram – Main Use Cases

Actor: Authorized Staff / Administrator

Add Student Record

View All Student Records

Search Student by ID or Name

Update Student Details

Delete Student Record

Generate or View Student Report (if implemented)

The actor interacts with the system through the user interface. The system processes each request and displays the result. Access control should be used if the application stores real student information.

7. Implementation

The Student Management System is implemented as an application for maintaining student records. The system is organized into a user interface, application logic, and a data storage component. The exact technologies and implementation details should be aligned with the working version submitted by the project team.

Main Functionalities

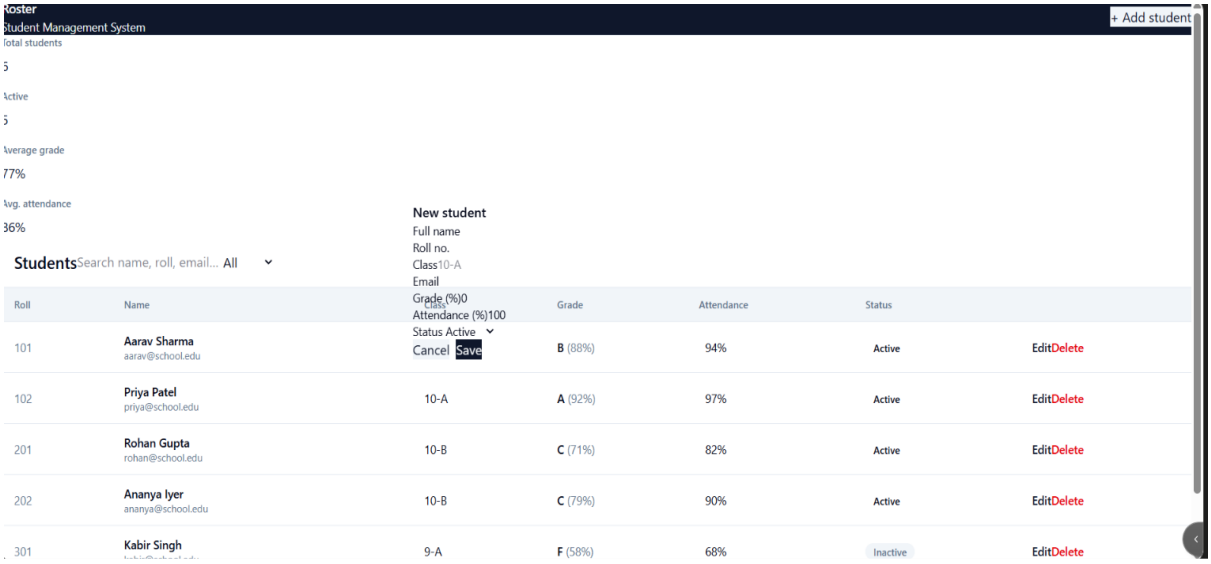
Add Student: Enter student ID or roll number, name, department, semester, and contact details.
View Students: Display the stored student records in an organized list or table.
Search Student: Find a record using a student ID, roll number, or name.
Update Student: Modify existing information when details change.
Delete Student: Remove a record when permitted and after confirming the action.
Validation: Check required fields and help prevent invalid or incomplete entries.

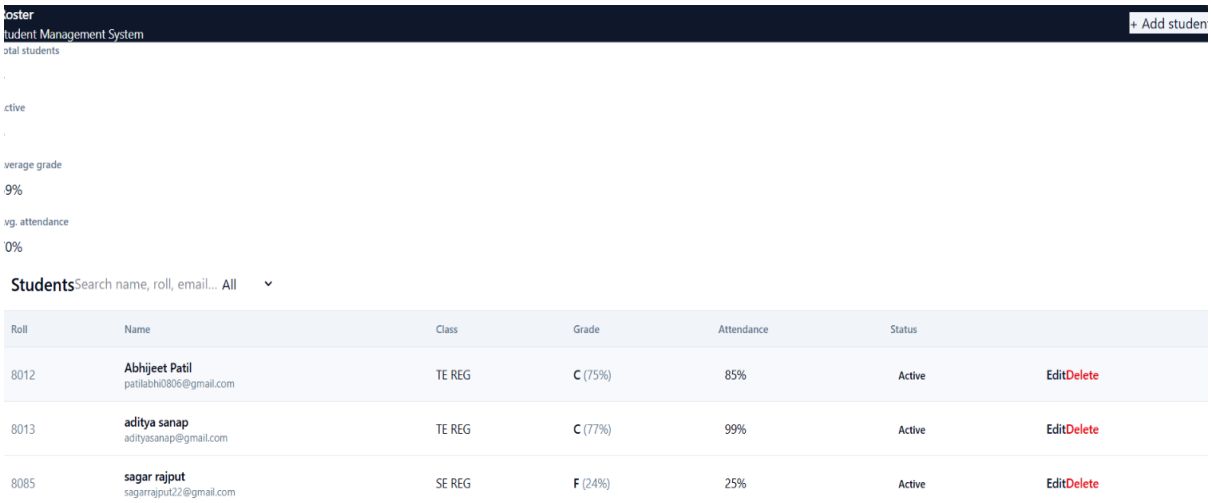
Typical Processing Flow

The user opens the application and selects an operation.
The user enters student details or a search key.
The application validates the input.
The application adds, retrieves, updates, or deletes the corresponding record.
A result or confirmation message is displayed to the user.
If the implementation uses an in-memory collection, records may be lost when the application closes. If it uses a database or file storage, describe the actual storage method here.

8. Outputs of the Mini-Project

The expected outputs of the Student Management System are:
Home screen or main menu for accessing student operations.
Student entry form for adding a new student.
Student list showing saved records.
Search result for a selected student.
Updated record after editing student details.
Confirmation message after a successful operation.
Validation or error message when required input is missing or invalid.
Insert screenshots from the actual running application in this section. Recommended screenshots are: (1) main screen, (2) add-student form, (3) student list, (4) search result, and (5) update or delete confirmation. Do not use sample screenshots that do not match the submitted application.





9. Skills Developed / Learning Out of This Mini-Project

Requirement analysis and identification of user needs.
Designing a simple and understandable user interface.
Programming application logic for record management.
Organizing student data and validating user input.
Testing add, view, search, update, and delete operations.
Debugging errors and improving application reliability.
Understanding the software development process from planning to testing.
Preparing technical documentation and coordinating project tasks.

10. Applications of This Mini-Project

Colleges and universities for maintaining student details.
Schools for managing basic student records.
Training institutes for storing learner information.
Departments for organizing student lists by semester or course.
Administrative offices for searching and updating student details.
As a foundation for extended systems with attendance, marks, fees, and report generation.

For real-world use, the application should include appropriate authentication, role-based access, data backups, and protection of personal information.

The Student Management System provides a structured way to maintain and manage student information. It supports common operations such as adding, viewing, searching, updating, and deleting student records, depending on the features implemented in the application. By organizing records in one system, it can reduce manual effort and make student information easier to retrieve.
This mini-project provides practical learning in requirement analysis, system design, implementation, data handling, and software testing. It also demonstrates how a software solution can address a routine administrative need in an educational institution. Future improvements may include secure login, attendance management, marks management, report generation, and database backup facilities.
Overall, the project serves as a useful foundation for understanding the development and maintenance of a student record management application.